WETLAND DEGRADATION ASSESSMENT

Executive Summary of Ecological & Hydrological Change Between Two Satellite Observations

OBSERVATION PERIOD	2023-08-14 → 2023-08-28	INTERVAL	14 days
WETLAND CONTEXT	Project-provided wetland boundary	RUN ID	run_20260929_094943
ANALYSIS FOOTPRINT	100.00% (100.00 ha)	SPATIAL REFERENCE	EPSG:32633

PRIMARY WETLAND ASSESSMENT

NO MEASURABLE WETLAND DEGRADATION DETECTED

Available multi-stream remote-sensing evidence demonstrates no systemic wetland degradation within the selected analysis area across the observation window.

INDICATOR	RESULT	AREA	PLAIN LANGUAGE INTERPRETATION
Hydro-Ecological Degradation	0.00%	0.00 ha	Concurrent canopy stress and surface water decline.
Vegetation Degradation	0.00%	0.00 ha	Vegetation canopy moving from healthier to more stressed tiers.
Vegetation Recovery / Regrowth	0.00%	0.00 ha	Vegetation canopy showing positive physiological recovery.
Surface Water Dynamics	N/A	0.00 ha	Net change in open surface moisture and water bodies.
Physical Wetland Loss	Not established	—	Conservative criterion: only claimed if verified by independent boundary evidence.

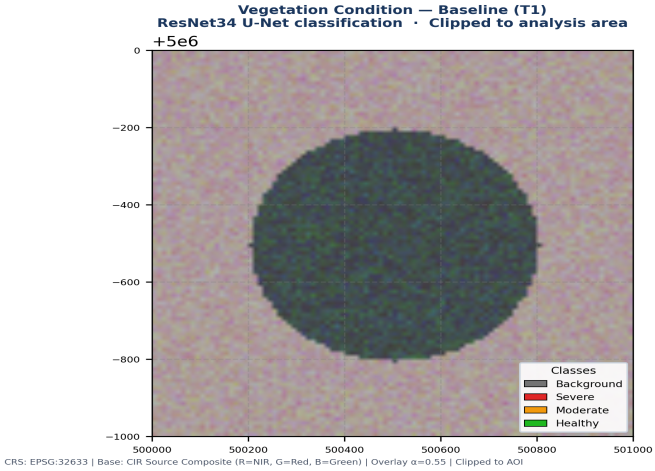
1. ANALYSIS OVERVIEW & METHODOLOGICAL FRAMEWORK

This study compares two satellite observations taken on **2023-08-14** (baseline) and **2023-08-28** (present), spanning an interval of **14 days**.
The objective is to identify whether measurable ecological degradation or surface water decline has occurred within the selected analysis area.
The evaluation examines three independent evidence streams: canopy health classification, continuous spectral change, and surface water dynamics.
All multi-spectral imagery is aligned to a common spatial reference grid. Radiometric normalization is applied to ensure cross-date reflectance comparability before computing spectral difference vectors and surface water boundaries across the effective analysis footprint (100.00 ha).

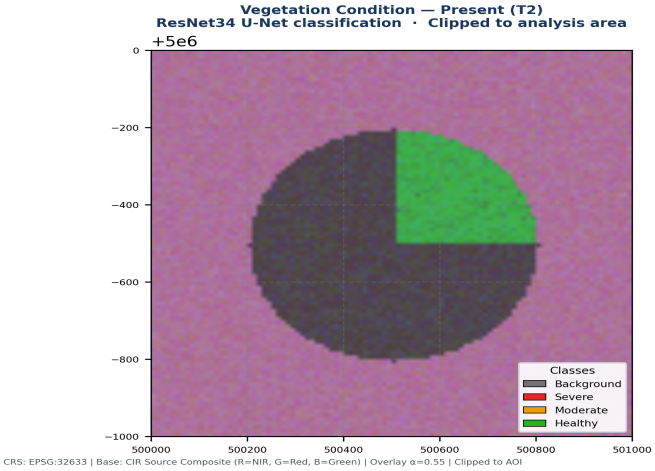
2. VEGETATION CONDITION AT T1 & T2

Vegetation canopy condition was classified independently at both observation dates into Healthy, Moderate Stress, Severe Stress, and Non-Vegetated Background:

CONDITION CATEGORY	T1 (2023-08-14)	T2 (2023-08-28)	NET SHIFT	PLAIN LANGUAGE INTERPRETATION
Healthy Vegetation	0.00%	0.00%	+0.0%	Dense, vigorous green canopy with high chlorophyll absorption.
Moderate Stress	0.00%	0.00%	+0.0%	Mild foliar moisture deficiency, early stress, or sparse cover.
Severe Stress	0.00%	0.00%	+0.0%	Marked foliage loss, dieback, or severe physiological stress.
Background / Non-Veg	100.00%	100.00%	+0.0%	Open water, bare soil, rock, or non-vegetated surfaces.



Baseline Vegetation Condition (T1: 2023-08-14)



Present Vegetation Condition (T2: 2023-08-28)

3. WHAT CHANGED? (VEGETATION TRANSITIONS)

Comparing T1 and T2 pixel-by-pixel reveals specific canopy trajectories: **0.0%** of the area showed degradation (canopy moving to a more stressed tier), **0.0%** demonstrated recovery, and the remainder remained stable.

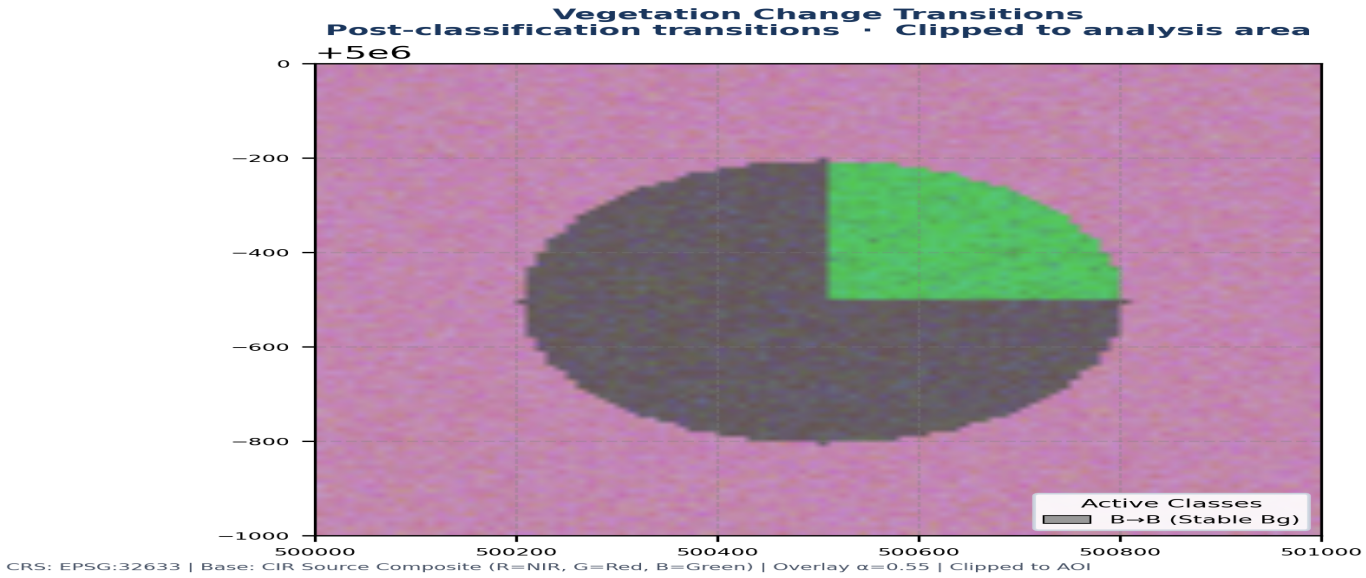


Figure 2: Vegetation Change Transitions Across the Analysis Area

4. SPECTRAL CHANGE (CONTINUOUS RADIOMETRIC SHIFTS)

Change Vector Analysis (CVA) measures the total magnitude of spectral shift across multiple wavelengths simultaneously. Unlike categorical classes, CVA captures subtle, continuous radiometric alterations. The average spectral displacement across the area was **0.4641**. Higher displacement indicates stronger physical alterations in canopy moisture or structure.

SPECTRAL INDICATOR	VALUE	DESCRIPTION
CVA Magnitude	0.4641	Overall Euclidean spectral displacement across Green, Red, RedEdge, NIR
Mean ΔNDVI	+0.3707	Difference in normalized vegetation greenness index
Mean ΔGNDVI	+0.1186	Difference in green chlorophyll absorption index
Mean ΔNDRE	-0.0134	Difference in red-edge foliar health index

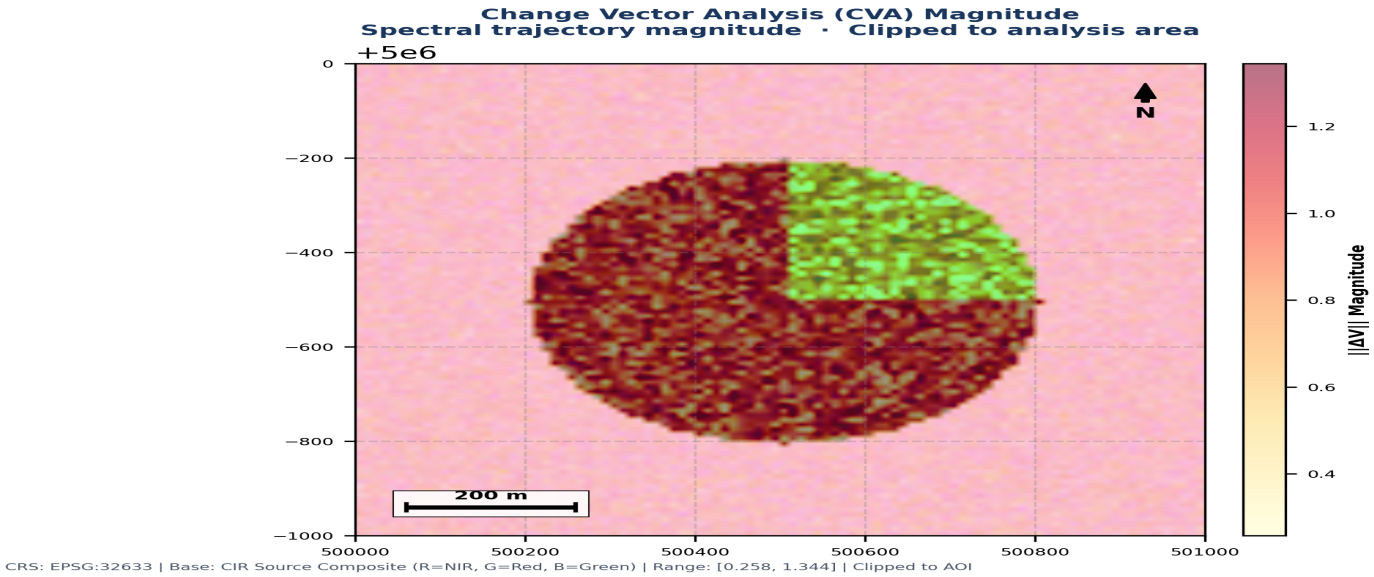


Figure 3: Change Vector Analysis (CVA) Magnitude Plate

5. HYDROLOGICAL EVIDENCE (SURFACE WATER DYNAMICS)

Surface-water dynamics indicate 6.72% expansion across the analyzed area. **Scientific Principle:** Vegetation condition change alone does NOT constitute physical wetland loss. A decline in greenness can stem from normal seasonal cycles or dry weather. True wetland degradation requires concurrent evidence of persistent surface water reduction or documented boundary loss.

HYDROLOGICAL METRIC	AREA (HA)	HYDROLOGICAL STATE DEFINITION
Baseline Water Extent (T1)	0.31 ha	Surface water area classified above NDWI threshold at T1 (2023-08-14)
Present Water Extent (T2)	7.03 ha	Surface water area classified above NDWI threshold at T2 (2023-08-28)
Surface Water Expansion (Gain)	6.72 ha	NDWI shifted above threshold (flooding / wetting)
Surface Water Contraction (Loss)	0.00 ha	NDWI shifted below threshold (drying / receding)

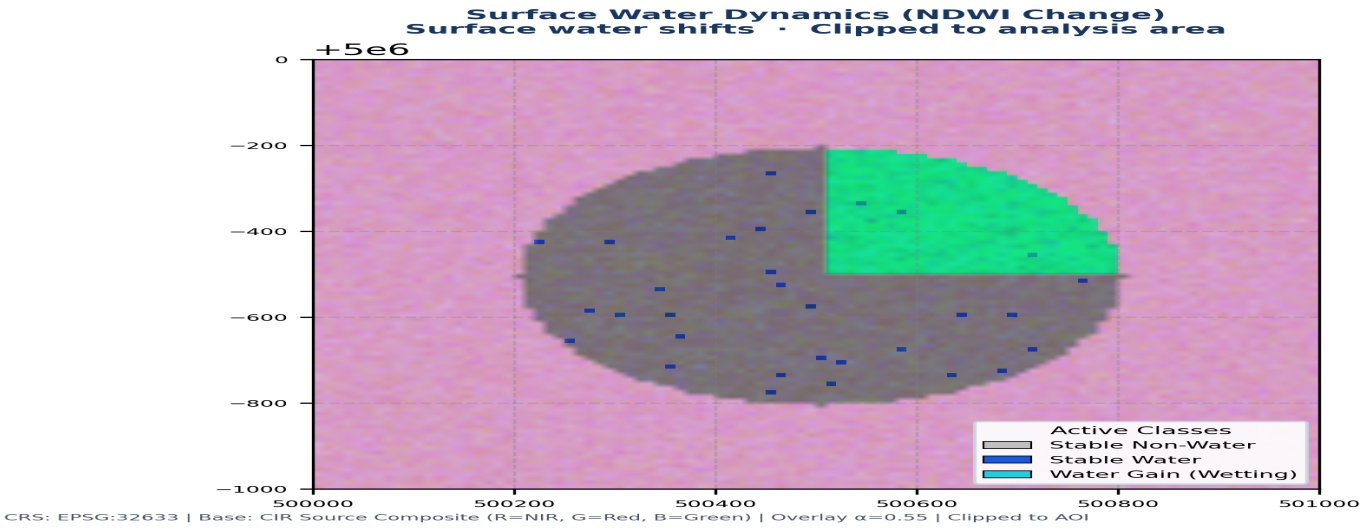


Figure 4: Surface Water Dynamics & Moisture Shifts

6. COMBINED EVIDENCE & MULTI-STREAM FUSION

The analysis is spatially bounded by the configured wetland boundary. All ecological condition indicators are evaluated within this defined area. To reach an objective conclusion, the GIS engine fuses vegetation transitions, continuous spectral displacement, and surface water dynamics into an integrated assessment:

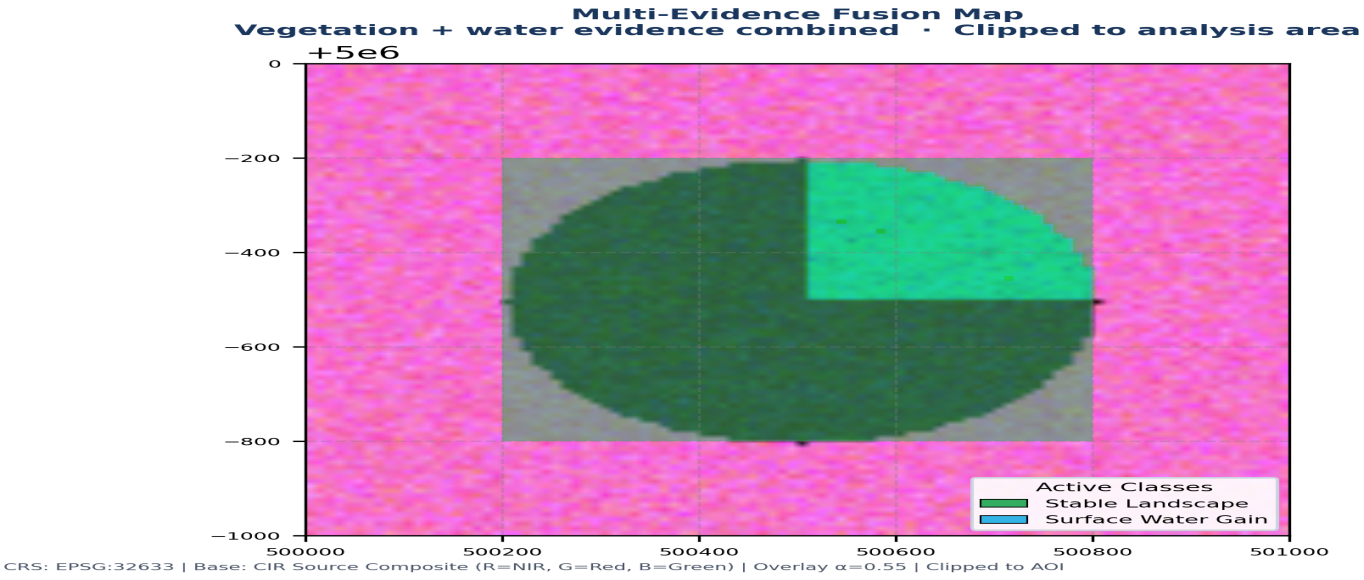


Figure 5: Multi-Evidence Fusion Map Combining Vegetation & Hydrological Data

7. FINAL WETLAND ASSESSMENT & KEY FINDINGS

Multi-evidence synthesis does not identify concurrent vegetation and hydrological decline sufficient to establish wetland degradation.

Physical wetland loss is not established. In accordance with strict scientific safeguards, optical vegetation change alone is not interpreted as permanent physical wetland loss.

- Vegetation canopy decline was observed across 0.0% of the analysis area.
- Vegetation recovery/regrowth was detected across 0.0% of the area.
- Surface-water dynamics: Surface-water dynamics indicate 6.72% expansion across the analyzed area.
- Multi-evidence synthesis: Available multi-stream remote-sensing evidence demonstrates no systemic wetland degradation within the selected analysis area across the observation window.
- Physical wetland depletion status: Not established.

8. LIMITATIONS & REPORT NOTES

Temporal Resolution: Analysis is based on two discrete overpass dates. Intermediate rainfall, flooding, or harvesting events between these dates cannot be directly captured.
Seasonal Phenology: Changes in vegetation vigor between different seasons may represent natural dormancy rather than anthropogenic degradation.
Ground Truth: High-priority degraded zones should be verified with field inspection before initiating statutory or enforcement actions.